Problem 1: Tip Calculator

What is the problem you are solving?What is the problem that you are addressing?

This program allows restaurant workers to easily find the tip and bill total. The user types in the amount of the bill and the percentage of the tip to be added, and the program calculates the amount of the tip and total amount to charge the customer.

STEP 2: Identify Inputs and Outputs

INPUTS:

- The total amount of money entered by the user.
- The percentage value the user enters for the Tip.
- Both inputs must be numbers (at least decimal numbers)

OUTPUTS:

- Tip amount
- Total amount with the tip.
- Results shown in dollar format ($)
- STEP 3: Work It By Hand

Example:

Bill amount = $45.80

Tip percentage = 15%

Steps:

Express 15% as a decimal:

15% = 0.15

Calculate the tip and add to the bill:

$45.80 × 0.15 = $6.87

Pay an extra fee on the bill:

$45.80 + $6.87 = $52.67

Final results to communicate to staff member:

“The tip amount is $6.87 and the total charge is $52.67.”

Problem 2: Late Fee Calculator

In this step, the problems are formulated.

STEP 1: Formulate the problems.

This program is designed for library staff to work out the overdue book fines. The librarian will enter the number of days that the book is late, and the program will calculate the amount of money that they will have to pay for the book being late.The librarian will enter in how many days the book is late and the program will calculate the amount of money the person needs to pay for the book being late up to a maximum of $10.00.

STEP 2: Identify Inputs and Outputs

INPUTS:

- Number of days the book is overdue
- The value of input type is an integer.

OUTPUTS:

- The amount of money that needs to be paid in fine.
- If the fine goes above $10.00, the program should display:
- “Maximum fine reached”

Special Cases:

- Fine should be $0.00 if the book is not late
- Fine cannot be more than $10.00
- Use your hands to work it.Use your hands to work it, STEP 3.

Case A: Book is 5 days late

My calculation:

5 × $0.50 = $2.50

Fine:

$2.50

It is 25 days late for book.Case B: Book is 25 days late.

My calculation:

25 × $0.50 = $12.50

Fine:

Maximum fine reached ($10.00)

Special note:

The normal fine is greater than $10.00 therefore the maximum fine limit is applicable.

Case C: Book is 0 days late meaning no late fees are due.

Fine:

$0.00